EDA — Reaction Time Experiment

A Course on Data Visualization · Group Activity

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Your task

Work in groups to prepare a **reproducible exploratory report** on reaction times and response accuracy in two experimental conditions. Your report should answer a clear question, support its conclusions with appropriate plots and descriptive statistics, and explain the limits of the evidence.

The activity has three steps:

1. **Explore and plan:** inspect the dataset, agree on an analysis plan, and identify the main issues that could affect interpretation.
2. **Prompt and verify:** use an AI assistant to help implement your plan, then run, check, and revise the proposed analysis.
3. **Discuss and compare:** compare your findings and analytical choices with those of other groups, then complete your report.

> **Plan before prompting.** Write down your group's initial questions and analysis choices before requesting an analysis from an AI assistant. Keep that initial plan so that you can explain what changed and why.

Study context and data collection

A visual decision experiment

The following study scenario is fictional and has been added for this teaching activity. It gives the dataset a concrete interpretation; it is not verified documentation of the data file's origin. The participant and trial counts below are the design described in the original activity and must be checked against the supplied data.

Imagine that a research team recruits 24 volunteers from a university to study how task difficulty relates to the speed and accuracy of visual decisions. On each trial, a participant sees two panels of dots on a computer screen and presses a key to indicate which panel contains more dots.

- **Condition A — easy:** the difference between the panels is large.
- **Condition B — hard:** the difference between the panels is small.

Each participant completes a short practice session, followed by 80 recorded trials: 40 in each condition. In this fictional protocol, the two conditions are randomly interleaved within each participant's session, and all participants use the same computer and response keys. Practice trials are excluded from the analysis file.

For every recorded trial, the software stores the participant identifier, the condition, the elapsed time from stimulus onset to the key press in milliseconds, and whether the response was correct. These records are exported to `RT_data.csv`.

This is a **within-participant experiment with repeated measurements**: every participant completes both conditions. The planned dataset therefore contains

$$24 \times 2 \times 40 = 1\,920 \text{ trial-level observations.}$$

> **A row is a trial, not a participant.** Repeated observations from the same person share that person's characteristics. The planned dataset contains 24 participants, not 1,920 independent participants. Comparisons between A and B should retain the pairing whenever participant-level results are used.

The fictional protocol motivates questions about task difficulty, but it does not establish the provenance or actual randomization of the supplied CSV. Your report should distinguish the teaching scenario from information that can be verified in the data. A convenience sample of volunteers also limits generalization to a broader population.

Variables

The original analysis uses the following columns. Check their types, observed values, and coding before using them.

Variable	Meaning	Role in the analysis
`Subject`	Participant identifier	Categorical identifier that links repeated trials and paired conditions
`Condition`	Task condition: A = easy, B = hard	Within-participant explanatory factor
`RT`	Reaction time, in milliseconds	Quantitative outcome; smaller values indicate faster responses
`Correct`	Response correctness, expected coding: 1 = correct, 0 = incorrect	Binary outcome; its mean gives the proportion correct when this coding is confirmed
`Correct_f`	A labelled version of response correctness	Categorical display variable; check that its labels agree with `Correct`

Do not assume that row order is trial order. If a trial index, timestamp, or block variable is available, inspect its meaning before using it to study practice or fatigue. Otherwise, those patterns cannot be assessed directly from the documented columns.

Get started

Packages and data file

This document uses `tidyverse`, `knitr`, and `kableExtra`. Install any missing packages once in the R console. Keep `RT_data.csv` in a folder named `data` next to this Rmd file, or edit `data_file` below to match its location. The CSV is an external input; this document does not generate replacement observations.

```r
library(tidyverse)
library(knitr)
library(kableExtra)
```

```r
data_file <- "data/RT_data.csv"

if (!file.exists(data_file)) {
  stop(
    "Data file not found. Put RT_data.csv in the data folder next to ",
    "this Rmd, or update data_file.",
    call. = FALSE
  )
}

dati <- read_csv(data_file, show_col_types = FALSE)

required_columns <- c("Subject", "Condition", "RT", "Correct", "Correct_f")
missing_columns <- setdiff(required_columns, names(dati))

if (length(missing_columns) > 0) {
  stop(
    "Required columns are missing: ",
    paste(missing_columns, collapse = ", "),
    call. = FALSE
  )
}
```

```r
glimpse(dati)
```

```
## Rows: 1,920
## Columns: 6
## $ Subject   <chr> "S01", "S01", "S01", "S01", "S01", "S01", "S01", "S01", "S01", "S01", ~
## $ Condition <chr> "A (Easy)", "A (Easy)", "A (Easy)", "A (Easy)", "A (Easy)", "A (Easy)"~
## $ Trial     <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,~
## $ Correct   <dbl> 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1~
## $ Correct_f <chr> "Correct", "Correct", "Incorrect", "Correct", "Correct", "Correct", "C~
## $ RT        <dbl> 548.5465, 456.2263, 623.4912, 573.0345, 417.7036, 486.7404, 335.4550, ~
```

```r
head(dati)
```

```
## # A tibble: 6 x 6
##   Subject Condition Trial Correct Correct_f    RT
##   <chr>   <chr>     <dbl>   <dbl> <chr>     <dbl>
## 1 S01     A (Easy)      1       1 Correct    549.
## 2 S01     A (Easy)      2       1 Correct    456.
## 3 S01     A (Easy)      3       0 Incorrect  623.
## 4 S01     A (Easy)      4       1 Correct    573.
## 5 S01     A (Easy)      5       1 Correct    418.
## 6 S01     A (Easy)      6       1 Correct    487.
```

Table 1: Table 1 · RT summary by Condition and Response Correctness

Condition	Response	N	Mean RT (ms)	Median RT (ms)	SD RT (ms)
A (Easy)	Correct	846	476.2	466.9	96.8
A (Easy)	Incorrect	114	546.1	534.1	103.8
B (Hard)	Correct	664	554.5	550.5	108.8
B (Hard)	Incorrect	296	612.2	603.9	113.7

Step 1 — Explore the dataset and plan the analysis

Complete this step as a group before requesting AI-generated analyses. Each member should contribute to the plan and be able to explain the group's decisions.

Inspect the data

Establish what is actually present in the file:

- Count the rows, distinct participants, and trials per participant and condition. Compare these counts with the planned design. Check whether every participant appears in both conditions.
- Inspect variable types, condition labels, and correctness coding. Cross-tabulate `Correct` and `Correct_f` to check consistency.
- Check missing values and inspect the range and distribution of `RT`. Flag non-positive values and investigate unusually short or long times.
- Consider whether apparently duplicated rows are genuine duplicates. Without a unique trial identifier, two identical recorded values may still correspond to different trials; do not delete them automatically.

Use small exploratory summaries and plots to understand the data. Keep a brief record of issues and decisions. Preserve the imported data and use new objects for any filtered or transformed versions.

Some simple descriptive statistics that can serve as a starting point:

```
# -- Overall summary ----------------------------------------
desc_overall <- dati %>%
  group_by(Condition, Correct_f) %>%
  summarise(
    N         = n(),
    Mean_RT   = mean(RT)   %>% round(1),
    Median_RT = median(RT) %>% round(1),
    SD_RT     = sd(RT)     %>% round(1),
    .groups   = "drop"
  )

desc_overall %>%
  kbl(caption = "Table 1 · RT summary by Condition and Response Correctness",
      col.names = c("Condition","Response","N",
                    "Mean RT (ms)","Median RT (ms)","SD RT (ms)")) %>%
  kable_styling(bootstrap_options = c("striped","hover","condensed"),
                full_width = FALSE) %>%
  row_spec(which(desc_overall$Correct_f == "Correct"),
           background = "#f0f7ff")
```

Table 2: Table 2 · Accuracy by Condition (subject-level means)

Condition	Mean Accuracy	SD
A (Easy)	0.881	0.050
B (Hard)	0.692	0.055

```
# -- Accuracy per subject x condition ---------------------
acc_subj <- dati %>%
  group_by(Subject, Condition) %>%
  summarise(Accuracy = mean(Correct), .groups = "drop")

acc_summary <- acc_subj %>%
  group_by(Condition) %>%
  summarise(
    Mean_Acc = mean(Accuracy) %>% round(3),
    SD_Acc   = sd(Accuracy)   %>% round(3),
    .groups  = "drop"
  )

acc_summary %>%
  kbl(caption = "Table 2 · Accuracy by Condition (subject-level means)",
      col.names = c("Condition", "Mean Accuracy", "SD")) %>%
  kable_styling(bootstrap_options = c("striped","hover","condensed"),
                full_width = FALSE)
```

Define your questions and planned outputs

Choose a main question about the relationship between condition and performance. Decide how reaction time and accuracy both contribute to your answer. For example, a faster response is not necessarily a better response if accuracy also changes.

Complete the following planning table **before Step 2**. Include at least one distribution plot, one plot that retains the within-participant comparison, and a compact table of descriptive statistics. Additional figures should answer a distinct question.

Question	Observations included and unit of summary	Planned plot or descriptive statistic	Reason for this choice	Main limitation
[Main question]	[All trials, correct trials, participant summaries, etc.]	[Your choice]	[What it reveals]	[What it may hide]
[Supporting question]	[Your choice]	[Your choice]	[Your justification]	[Your concern]
[Supporting question]	[Your choice]	[Your choice]	[Your justification]	[Your concern]

Step 1 output: your initial analysis plan, a short data-quality record, and a list of decisions that need particular care. Keep this version even if the analysis changes later.

Step 2 — Prompt the analysis, then verify it

Turn your plan into a prompt

Use your Step 1 plan to request specific analytical work from an AI assistant. Provide the variable definitions, verified coding, study structure, and the outputs you want. An assistant needs the relevant data to calculate results; a description of the dataset alone is not enough. If it cannot access the CSV, request code to run locally and do not accept invented numerical findings.

Run, check, and revise

Execute the proposed code in R and inspect the outputs. A plausible-looking chart or fluent explanation is not sufficient evidence that an analysis is correct.

- Check that the code implements your planned grouping, filters, and denominators. Verify that paired values belong to the same participant.
- Recalculate at least one reaction-time summary and one accuracy value directly for a single participant and condition. Compare them with the generated table or plot.
- Inspect warnings, missing results, and any participants lost during filtering or joining. Keep enough intermediate summaries to explain them.
- Verify labels, units, scales, legend entries, and the numbers cited in the written interpretation against the actual output.
- Review suggestions that change your plan. Accept, revise, or reject them with a reason, and use follow-up prompts when a correction is needed.

Keep the main prompt and the important follow-up prompts. Maintain a short decision log; it should show what you checked, not merely list AI responses.

AI suggestion or output	Check performed	Decision and reason	Consequence for the report
[Suggestion or output]	[Your verification]	[Accepted, revised, or rejected; why]	[What changed, if anything]

> **Step 2 output:** working R code, verified figures and tables, a draft interpretation, and a prompt and decision log. Your group is responsible for every result and statement retained in the report.

Step 3 — Discuss and compare results

Share your draft findings with another group. Explain your main question, one key figure, one descriptive table, and one decision that materially affects interpretation.

Compare both the conclusions and the choices behind them:

- Did the groups analyse the same trials and use the same unit of summary?
- Did reaction time and accuracy support the same description of performance?
- Were the apparent differences similar across participants, or did participant-level plots reveal important variation?
- If numerical results differ, can the difference be traced to filters, missing values, denominators, weighting, or the choice of summary?
- Did a particular graphical choice make a pattern clearer or obscure it?

- Which AI suggestions needed correction, and how were problems detected?

When results disagree, reproduce at least one disputed calculation using a shared definition before interpreting the disagreement. Different analyses may answer different questions; explain this when it applies.

Revise your report after the discussion. State one useful point of agreement, one difference between the analyses, and one improvement you made. If no revision is needed, explain which checks supported retaining your approach.

> **Step 3 output:** a short comparison of the groups' approaches and the final, revised report.

The report to submit

Submit one **Rmd source file and its knitted HTML report per group**. Include the group members' names. Keep the data path reproducible and state which supplied data file you used. The report must knit in a fresh R session with the CSV and required packages available; it should not depend on objects created manually in the Console.

Use the following structure. Replace the activity instructions and placeholders with your own analysis and explanations in the submission.

Section	Required content
Research question and context	Main question, brief study description, and explicit identification of the fictional scenario
Data and analysis decisions	Observed data dimensions and coding, repeated-measures structure, data-quality findings, inclusion rules, denominators, and choice of summaries
Results	At least one distribution plot, one plot retaining the within-participant comparison, and a concise descriptive table; cover both reaction time and accuracy
Interpretation and limitations	Answers supported by actual results, relevant variability, and limits arising from the data and analytical choices
Comparison and revision	What comparison with another group revealed and how the final report changed
Reproducibility and AI appendix	Initial plan, main and relevant follow-up prompts, decision log, reproducible code, and a brief statement of AI assistance

Every figure and table needs a caption identifying what is summarized, its units, and the observations included. Add a short interpretation in your own words. Use a small number of purposeful outputs rather than a catalogue of charts, and avoid describing a difference as statistically significant without an appropriate inferential analysis.

The report will be assessed for the clarity of its questions, suitability of its plots and summaries, treatment of repeated measurements and data-quality issues, quality of interpretation, reproducibility, and evidence that AI-generated work was checked critically.

An AI-assistance statement can follow this template, adapted to the work actually performed:

AI assistance. We used [tool/model, if available; date] to assist with [specific tasks]. We checked [specific code, calculations, and interpretations] and revised [main issues]. The group takes responsibility for the submitted analysis and conclusions.